

StyleMate -
An AI-Powered Virtual Assistant for Personalized Clothing
Selection Using 3D Body Modeling
(Accessory Recommendation & Customization System)
25-26J-302

Project Proposal Report
Vishnuyan.T - IT22208194

B.Sc. (Hons) in Information Technology Specializing in Information
Technology

Department of Computer Science & Software Engineering

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Declaration

I say that this is my own work, and this project does not use any work that was given before for a degree or diploma at any other university or college without saying where it came from. As far as I know, it does not have any work that was published or written by another person before, except where I have said who wrote it in the text.

Name	Student ID	Signature
Vishnuyan.T	IT22208194	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....

Signature of the supervisor:

(Ms. Anjalie Gamage)

Date:.....

.....

Signature of the Co-supervisor:

(Prof. Samantha Rajapaksha)

Date:.....

Abstract

The Accessory Recommendation & Customization System introduces an innovative component within virtual dressing room technology, designed to enhance personalized fashion experiences. By leveraging computer vision, the system processes user-uploaded full-body images to extract critical styling attributes, including body shape, face shape, dress color, neckline style, and skin tone. These attributes drive a recommendation engine that applies fashion styling rules to suggest complementary accessories, such as rope chain necklaces for V-neck dresses, ensuring precise and context-aware styling. Beyond automated recommendations, the system incorporates a customization feature, enabling users to request specific accessory designs, such as heart-patterned necklaces, which are either generated or retrieved based on user preferences. This dual approach integrates data-driven style intelligence with user-driven creativity, addressing limitations in existing systems that lack personalization and customization. The proposed system aims to deliver an inclusive, user-centric virtual fashion solution, advancing the accessibility and expressiveness of digital styling for diverse users.

Table of Contents

Abstract.....	4
List of Figure.....	6
List of Table	6
List of Abbreviations.....	7
1.0 INTRODUCTION.....	8
1.1 Background and Literature.....	8
1.2 Research Gap.....	9
1.3 Research Problem	10
2.0 Objectives.....	11
2.1 Main Objective	11
2.2 Specific Objectives.....	11
3.0 METHODOLOGY.....	12
3.1 Requirement Gathering.....	12
3.1.1 Past Research Analysis.....	13
3.1.2 Tools & Technology.....	14
3.2 Project Requirements.....	15
3.2.2 Non-Functional Requirements.....	15
3.3 Testing.....	16
3.4 Timeline.....	17
4.0 Risk Management Plan	18
5.0 COMMERCIALIZATION.....	20
6.0 BUDGET.....	21
7.0 SUMMARY	22
References.....	23

List of Figure

Figure 1: System Diagram	16
Figure 2: System Overview	17
Figure 3: Gantt Chart for Project Schedule Management.....	17

List of Table

Table 1: List of abbreviations	7
Table 2: Risk Management Plan	19
Table 3: Cost Management Plan	21

List of Abbreviations

Key words	Meaning
CV	Computer Vision
ML	Machine Learning
UI	User Interface
API	Application Programming Interface
VS Code	Visual Studio Code

Table 1: List of abbreviations

1.0 INTRODUCTION

1.1 Background and Literature

The concept of a virtual dressing room is driven by the increasing demand for personalized, online shopping experiences that integrate convenience, technology, and inclusivity. Traditionally, consumers relied on physical try-ons to evaluate clothing fit and accessory compatibility. The shift to online shopping has spurred the development of virtual platforms to replicate this experience. Virtual dressing rooms employ artificial intelligence (AI), machine learning (ML), and computer vision to generate digital avatars or 2D models, allowing users to visualize clothing and accessories before purchasing. Early systems were limited to basic 2D representations, often using static images or simplistic overlays, lacking the ability to account for wearer-specific attributes or personalized styling.

Recent advancements in virtual fashion have focused on enhancing personalization through AI-driven systems. A framework for personalized outfit recommendations uses deep learning to analyze user images and extract attributes like clothing style and body shape, employing convolutional neural networks (CNNs) to suggest style-compatible outfits [1]. While effective for clothing ensembles, this approach does not address accessory-specific recommendations or user-driven customization, which are essential for a comprehensive virtual styling solution.

Personalized recommendation systems have also evolved, with models integrating visual features from product images with user preferences to suggest complementary fashion items [2]. This approach excels in general fashion recommendations but does not incorporate wearer-specific attributes like face shape, neckline style, or skin tone, nor does it support user-defined customization, limiting its applicability to accessory-focused systems like the proposed Accessory Recommendation & Customization System.

Customization in virtual fashion has gained momentum through generative AI. A framework for generating custom clothing designs based on user text inputs uses cycle-consistent generative adversarial networks (GANs) to create novel patterns [3]. However, this system focuses on apparel and lacks real-time attribute extraction from user images, which is critical for context-aware accessory recommendations. Similarly, a virtual styling system analyzes user-uploaded images to recommend accessories based on clothing attributes [4]. While effective, it lacks a customization module, restricting user control over design preferences.

The Accessory Recommendation & Customization System addresses these gaps by integrating computer vision to extract styling attributes (body shape, face shape, dress color, neckline style, skin tone) from user-uploaded images, a rule-based recommendation engine to suggest accessories (e.g., rope chain necklaces for V-neck dresses), and a customization module allowing users to request specific designs (e.g., heart-patterned necklaces). This dual approach, inspired by [3], enhances user agency through data-driven recommendations and personalized design generation or retrieval. The importance of inclusive, user-centric fashion systems is emphasized, noting that adaptability to diverse user profiles is critical for modern virtual fashion platforms. The proposed system aligns with this trend, offering a comprehensive solution that bridges automated styling expertise with individual creativity, advancing the inclusivity and expressiveness of virtual dressing rooms.

1.2 Research Gap

The Accessory Recommendation & Customization System integrates computer vision for extracting styling attributes (body shape, face shape, dress color, neckline style, skin tone), a rule-based recommendation engine, and a user-driven customization module to enhance virtual dressing rooms. Despite advancements in virtual fashion, significant gaps persist in accessory-focused recommendations, wearer-specific attribute integration, and user-controlled customization, limiting comprehensive and inclusive styling solutions.

Current virtual fashion systems prioritize clothing over accessories, neglecting their role in completing a look. For instance, a system for fashion compatibility recommends clothing items based on visual and contextual features but overlooks accessories tied to wearer attributes like neckline style or face shape [5]. Similarly, personalized outfit generation systems leverage visual features and user preferences but fail to incorporate detailed attributes such as body shape or skin tone, reducing the precision and inclusivity of accessory recommendations, such as matching rope chain necklaces to V-neck dresses [6].

Generative AI enables custom apparel designs from text prompts but lacks real-time attribute extraction from user images, critical for context-aware accessory recommendations [7]. Existing systems also rarely allow users to request or modify accessory designs, limiting creative expression. The proposed system addresses these gaps by combining real-time attribute extraction, precise accessory recommendations, and user-specified design generation or retrieval, offering a novel, inclusive solution that bridges automated styling with individual creativity in virtual fashion.

1.3 Research Problem

The rise of online shopping has increased the demand for virtual dressing room technologies that replicate the personalized experience of physical try-ons. However, existing systems fall short in delivering comprehensive and inclusive styling solutions, particularly for accessories. Most virtual fashion platforms focus on clothing recommendations, overlooking the critical role of accessories in enhancing a user's appearance. These systems often fail to integrate wearer-specific attributes, such as body shape, face shape, dress color, neckline style, and skin tone, which are essential for precise and context-aware accessory recommendations. For example, recommending a rope chain necklace to complement a V-neck dress requires analyzing specific attributes that current systems do not adequately address. Additionally, there is a lack of user-driven customization options, limiting individuals' ability to request specific accessory designs, such as heart-patterned necklaces, to express personal creativity. This results in generic, apparel-centric recommendations that do not fully cater to diverse user profiles or preferences. The absence of an integrated system combining real-time attribute extraction, targeted accessory recommendations, and user-controlled customization creates a significant challenge in providing a personalized, inclusive, and expressive virtual styling experience. The Accessory Recommendation & Customization System aims to address this problem by leveraging computer vision to extract styling attributes, employing a rule-based engine for precise accessory suggestions, and offering a customization module to empower users, thereby bridging the gap between automated styling and individual creativity in virtual fashion.

2.0 Objectives

2.1 Main Objective

The primary objective of the Accessory Recommendation & Customization System is to develop an intelligent virtual dressing room component that delivers personalized, inclusive, and creative accessory styling solutions. By integrating computer vision to extract wearer-specific attributes (body shape, face shape, dress color, neckline style, and skin tone) from user-uploaded images, the system aims to provide precise, context-aware accessory recommendations, such as suggesting rope chain necklaces for V-neck dresses. Additionally, it seeks to empower users through a customization module that enables them to request specific accessory designs, like heart-patterned necklaces, which are generated or retrieved based on their preferences. This dual approach combines automated, data-driven styling with user-driven creativity to create a comprehensive, accessible virtual fashion experience that addresses diverse user profiles and enhances individual expression.

2.2 Specific Objectives

The Accessory Recommendation & Customization System aims to achieve the following specific objectives to deliver a personalized, inclusive, and creative virtual styling experience:

Develop a Computer Vision Module for Attribute Extraction: Implement a robust computer vision system to accurately extract wearer-specific attributes, including body shape, face shape, dress color, neckline style, and skin tone, from user-uploaded full-body images, ensuring precise input for the recommendation engine.

Design a Rule-Based Recommendation Engine for Accessories: Create an intelligent recommendation engine that applies fashion styling rules to suggest complementary accessories, such as recommending rope chain necklaces for V-neck dresses, based on extracted attributes to enhance aesthetic compatibility and inclusivity across diverse user profiles.

Implement a User-Driven Customization Module: Develop a feature that allows users to request specific accessory designs or patterns, such as heart-patterned necklaces, by generating or retrieving these items from a catalog, enabling personalized creative expression within the virtual dressing room.

Ensure Inclusivity and Accessibility: Optimize the system to accommodate diverse user profiles by accounting for variations in body shape, skin tone, and cultural styling preferences, ensuring fair and relevant recommendations and customization options.

Integrate and Evaluate System Performance: Combine the computer vision, recommendation, and customization modules into a cohesive system and evaluate its effectiveness through user studies to measure accuracy, user satisfaction, and inclusivity in delivering personalized virtual fashion experiences.

3.0 METHODOLOGY

3.1 Requirement Gathering

To develop the Accessory Recommendation & Customization System, a comprehensive requirement gathering process is essential to ensure the system meets user needs, aligns with technical feasibility, and supports personalized virtual styling. This process involves stakeholder interviews, surveys, user observations, focus groups, literature and market analysis, and prototyping with feedback. Fashion designers, and e-commerce developers will identify desired features, such as recommending rope chain necklaces for V-neck dresses or enabling custom heart-patterned necklace designs. Surveys will quantify user preferences for accessory types and styling attributes like body shape and skin tone, ensuring inclusivity. Observing users on existing platforms will reveal usability issues, while focus groups with diverse participants will explore customization needs and cultural preferences. Literature and market analysis will inform technical requirements, such as computer vision for attribute extraction and generative AI for customization. Initial prototypes will be tested to validate and refine requirements, ensuring the system delivers precise, inclusive accessory recommendations and user-driven customization for a seamless virtual fashion experience.

3.1.1 Past Research Analysis

Past research in virtual fashion and recommendation systems has laid a foundation for the Accessory Recommendation & Customization System, which integrates computer vision, rule-based recommendations, and user-driven customization. However, existing studies reveal significant limitations in accessory-focused recommendations, wearer-specific attribute integration, and user-controlled design, highlighting the need for a more comprehensive and inclusive virtual styling solution.

Early virtual dressing room systems relied on basic 2D representations, overlaying static clothing images onto user models, but lacked personalization and accessory focus. Advancements in computer vision enabled more sophisticated systems. For instance, a deep learning-based approach used bidirectional LSTMs to recommend clothing based on visual compatibility, achieving improved outfit coherence [5]. However, this system primarily addressed apparel, neglecting accessories and wearer-specific attributes like body shape or neckline style, limiting its applicability to precise accessory recommendations.

Personalized recommendation systems have progressed by incorporating visual features and user preferences. A notable system developed for Alibaba iFashion generated personalized outfits by combining visual and contextual data, enhancing recommendation relevance [6]. Despite its success, it overlooked detailed attributes such as face shape or skin tone and lacked mechanisms for accessory-specific suggestions or user-driven customization, resulting in generic styling outputs that fail to address diverse user needs.

Generative AI has advanced virtual fashion customization, with diffusion-based models enabling text-guided clothing design generation, producing high-quality, user-specified patterns [7]. While innovative, these models focus on apparel and do not integrate real-time attribute extraction from user images, critical for context-aware accessory recommendations. Additionally, they lack interactive customization for accessories, restricting user creative control.

These studies collectively demonstrate progress in virtual fashion but highlight gaps in accessory-focused systems, integration of wearer-specific attributes, and user-driven customization. The Accessory Recommendation & Customization System builds on these foundations by leveraging computer vision to extract attributes (body shape, face shape, dress color, neckline style, skin tone), employing a rule-based engine for precise accessory suggestions (e.g., rope chain necklaces for V-neck dresses), and offering a customization module for user-requested designs (e.g., heart-patterned necklaces). This approach addresses the limitations of past research, delivering a novel, inclusive, and expressive virtual styling solution.

3.1.2 Tools & Technology

The Accessory Recommendation & Customization System leverages a focused set of tools and technologies to enable computer vision-based attribute extraction, rule-based accessory recommendations, and user-driven customization in a virtual dressing room.

Programming Language: Python – For developing computer vision, recommendation logic, and backend processing.

Frameworks and Libraries:

TensorFlow – For building deep learning models (e.g., CNNs) to extract attributes (body shape, face shape, dress color, neckline style, skin tone).

OpenCV – For image processing and feature extraction.

PyTorch – For generative AI models to create custom accessory designs (e.g., heart-patterned necklaces).

Database: MongoDB – To store user profiles, styling attributes, and accessory catalogs.

Web Development:

React JS – For interactive front-end interface.

Material-UI (MUI) – For designing UI components.

UI Design: Figma – For prototyping user interfaces.

Version Control: Git (GitHub) – For code management and collaboration.

Testing Tools:

Pytest – For Python unit testing.

Postman – For API testing.

Deployment Tools:

Docker – For containerization.

Kubernetes – For orchestration and scalability.

Project Management: Trello – For task organization and progress tracking.

3.2 Project Requirements

3.2.1 Functional Requirements

The functional requirements for the proposed model are as follows:

The Accessory Recommendation & Customization System delivers personalized virtual styling through computer vision, rule-based recommendations, and user-driven customization. Key functional requirements include:

Image Upload: Allow users to upload full-body images via a web interface.

Attribute Extraction: Extract body shape, face shape, dress color, neckline style, and skin tone using computer vision with 85% accuracy.

Accessory Recommendation: Suggest at least three inclusive accessory options (e.g., rope chain necklaces for V-neck dresses) based on attributes.

Customization: Enable users to request specific designs (e.g., heart-patterned necklaces) via text, generating or retrieving them.

User Interface: Provide a responsive React JS and Material-UI interface for real-time visualization of recommendations and custom designs.

Data Management: Store user profiles and preferences in MongoDB, ensuring data privacy.

Integration: Combine computer vision (TensorFlow, OpenCV), recommendation, and customization modules with API support.

Performance: Process images and deliver recommendations

3.2.2 Non-Functional Requirements

The non-functional needs that were prioritized throughout the creation of the suggested model are listed below.

- Usability
- Reliability
- Availability
- Accuracy
- Performance

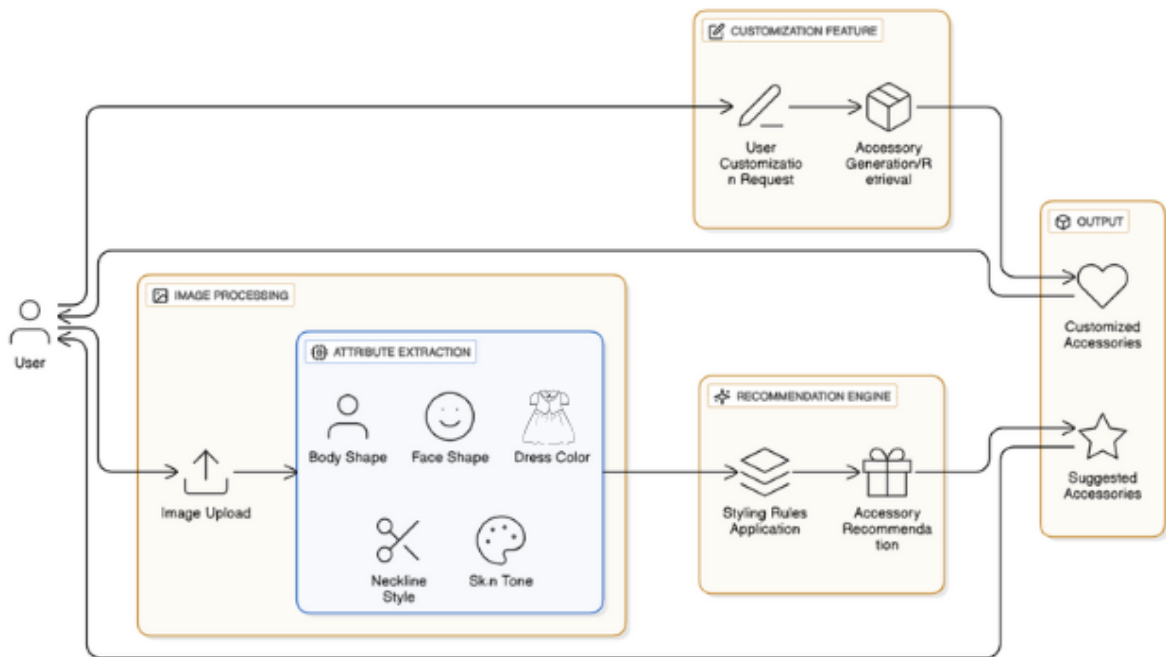


Figure 1: System Diagram

3.3 Testing

The Accessory Recommendation & Customization System will undergo thorough testing to ensure accurate attribute extraction, precise accessory recommendations, and seamless user-driven customization within the virtual dressing room. Unit testing with Pytest will validate individual components, such as computer vision algorithms (TensorFlow, OpenCV) for extracting attributes like body shape and dress color, targeting 85% accuracy across diverse images. Integration testing will confirm the cohesive operation of the computer vision, recommendation engine, and customization modules, ensuring API endpoints (tested via Postman) deliver recommendations like rope chain necklaces for V-neck dresses. User acceptance testing will involve diverse participants evaluating the React JS and Material-UI interface for usability, inclusivity, and real-time visualization of customized designs, such as heart-patterned necklaces. Performance testing will verify system responsiveness under typical load, while stress testing will assess stability during peak usage. Test cases will ensure inclusivity across skin tones and cultural preferences, with results analyzed to enhance system reliability and user satisfaction.

Accessory Recommendation System

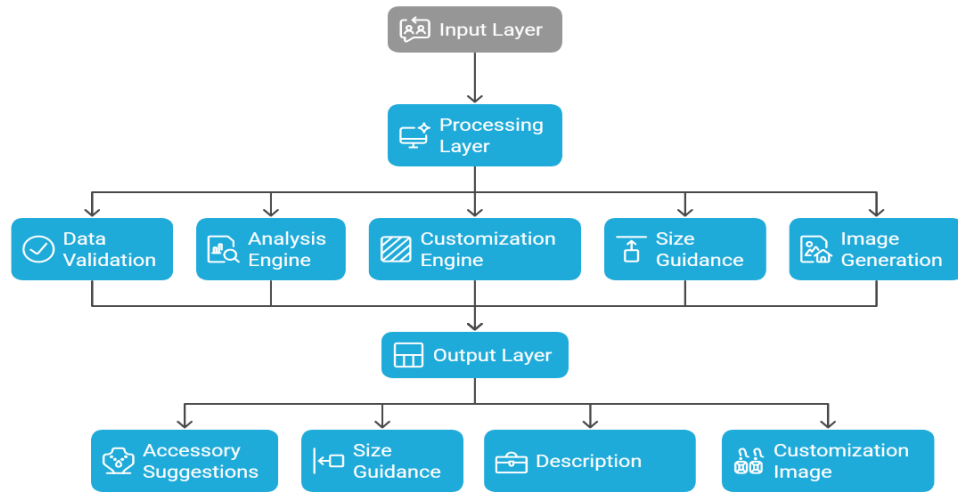


Figure 2: System Overview

3.4 Timeline

The proposed timeline for the project is as follows.

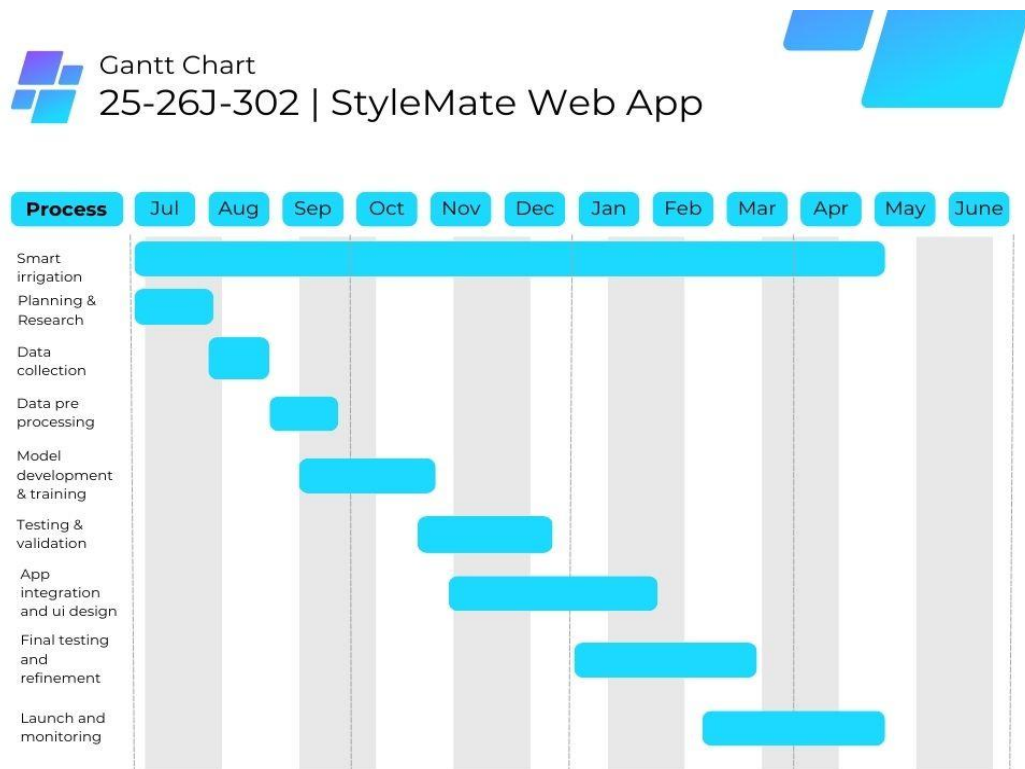


Figure 3: Gantt Chart for Project Schedule Management

4.0 Risk Management Plan

The Accessory Recommendation & Customization System, integrating computer vision, rule-based recommendations, and user-driven customization for virtual dressing rooms, requires a comprehensive risk management plan to address potential challenges during development and deployment. The table below outlines identified risks, their descriptions, probability, impact, priority, ownership, mitigation strategies, and contingency plans to ensure successful project execution and system reliability.

Risk	Description	Probability	Impact	Priority	Owner	Mitigation Strategy	Contingency Plan
Inaccurate Attribute Extraction	Computer vision fails to accurately extract attributes (e.g., body shape, skin tone) due to diverse image quality or lighting conditions.	Medium (40%)	High	High	Computer Vision Lead	Use robust preprocessing (OpenCV) and train models (TensorFlow) on diverse datasets with varied lighting and backgrounds. Conduct rigorous unit testing with Pytest.	Implement fallback manual attribute input option for users to specify attributes if extraction fails.
Ineffective Recommendations	Rule-based engine suggests irrelevant accessories (e.g. mismatched necklaces) due to incomplete styling rules or biased data.	Medium (30%)	High	High	Recommendation System Lead	Develop comprehensive fashion styling rules validated by experts. Test recommendations on diverse user profiles to ensure inclusivity.	Allow users to provide feedback on recommendations to refine rules iteratively.
Customization Module Failure	System fails to generate or retrieve user	Low (20%)	Medium	Medium	Customization Lead	Use PyTorch based generative AI trained on diverse	Provide a default set of predesigned accessoires

	requested designs (e.g. heart-patterned necklaces) due to limited catalog or generative AI issues.					accessory designs and maintain an extensive MongoDB catalog. Test with varied user inputs.	as alternatives if generation fails
Data Privacy Breach	Sensitive user data (e.g. images, preferences) is exposed during processing or storage.	Low (15%)	Critical	High	Security Lead	Implement anonymization for images and encrypt data in MongoDB. Conduct security testing to ensure compliance with data protection standards.	Activate immediate system lockdown and notify users if a breach is detected with a backup data restoration plan.
System Scalability Issues	System cannot handle high user loads causing delays or crashes.	Medium (35%)	High	High	DevOps Lead	Use Docker for containerization and Kubernetes for orchestration. Perform stress testing to validate scalability.	Deploy additional server instances and optimize load balancing to handle unexpected traffic spikes.
User Interface Usability Issues	React JS and Material UI interface is unintuitive, hindering user interaction with recommendations or customization.	Low (25%)	Medium	Medium	Front-End Lead	Design intuitive UI with Figma prototypes and conduct user acceptance testing with diverse participants.	Provide user tutorials and tooltips within the interface to guide navigation and feature usage.

Table 2: Risk Management Plan

5.0 COMMERCIALIZATION

Target Market: Sri Lanka's fashion e-commerce market is valued at LKR 19.8 billion (2024), with rapid growth across Asia. Wrong purchases & return rates (~30%) highlight a clear pain point.

Business Model: Freemium web app with premium subscription (LKR 500/month) for personalized style insights & wardrobe features. Partnerships with online retailers and fashion brands generate additional revenue via affiliate links.

Cost Recovery: Development cost (~LKR 3.5 million) can be recovered within 12–15 months by targeting even 10,000 active users in Sri Lanka (and scaling to South Asia).

Unique Selling Points (USP):

First explainable fashion recommender fusing body + skin analysis.

Inclusive of Asian body types & skin tones, unlike global systems.

Complete ecosystem (outfits, colours, accessories, occasions).

Partnership Opportunities:

Local fashion retailers

Online platforms

Boutique designers offering personalized services.

6.0 BUDGET

Category	Description	Estimated Cost (LKR)
Hosting & Cloud	Basic cloud server / GPU rental for 6 months	30,000 .00
Domain & Web Hosting	Domain (stylemate.lk) + basic web hosting	10,000.00
Publication	Conference/journal fees + printing/binding	12,500 .00
Deployment	Cloud deployment of prototype (3–4 months)	15,000.00
Marketing (Optional)	Posters, brochures, small-scale promotions	20,000.00
Total Estimated		87,500.00

Table 3: Cost Management Plan

7.0 SUMMARY

The Accessory Recommendation & Customization System serves as an intelligent component within virtual dressing room technology that processes user-uploaded full-body images to extract critical styling attributes including body shape, face shape, dress color, neckline style, and skin tone. This extracted data feeds into a recommendation engine that applies fashion styling rules to suggest complementary accessories, such as recommending rope chain necklaces for users wearing V-neck dresses. The system extends beyond automated recommendations by incorporating a customization feature that allows users to request specific accessory designs or patterns, with the system generating or retrieving the requested items like heart-patterned necklaces based on user preferences. This dual approach combines computer vision-powered attribute recognition with style intelligence algorithms to deliver both data-driven fashion recommendations and personalized accessory customization, creating a comprehensive solution that bridges automated styling expertise with individual user creativity and preferences in the virtual fashion space.

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